

The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Fire-Station System Specification

System Architecture Invariant Manifest

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1 Introduction: The Unified Search-and-Rescue Plot

We define the complete search-and-rescue architecture for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS). The terminal objective of the system is the minimization of target coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in the physical extraction of the lost asset. Rather than relying on statistical dynamic programming, the system decomposes into eleven distinct, interacting parts ($c \in C$) coordinated by an Apex Neurosymbolic Engine.

2 The Eleven Core Structural Subsystems

2.1 The Fire-Station Dispatcher

The centralized apex root node that interfaces with the Master Functor Dispatcher (Tree #0). It accepts high-level natural language problem descriptions, checks database axioms, and triggers fleet deployment vectors.

2.2 The 50 Mobile Drone Platforms (UAVs)

A distributed fleet of autonomous double-integrator kinematic plants. Each vehicle hosts a local instance of the LISP DSL compiler and a compact neuro-symbolic engine (utilizing an LLM coding agent) to fill programmatic policy stencils (holes) on-the-fly based on immediate feedback.

2.3 The Lagrangian Weather Sensor Array

Co-moving atmospheric fluid probes mounted on each airborne platform. These probes measure local velocity vectors, feeding a Discrete Wavelet Transform (DWT) to filter high-frequency turbulence and expose scale-invariant advection trends without statistical state space estimation.

2.4 The Down-View Visual Sensor

An optical targeting payload dedicated to scanning non-convex ravine coordinates. Upon target matches, it logs the mandatory discovery passes and initializes target packet tracking.

2.5 The Communications Mesh Graph ($\mathcal{G}_C$)

A fragile topology governed by a strict 30-mile line-of-sight constraint. It maintains instantaneous mesh connectivity without store-and-forward defaults, requiring non-zero algebraic connectivity to prevent network torso fracturing ($\beta_0 > 1$).

2.6 The Computational Tasking Graph ($\mathcal{G}_{Comp}$)

A network tracking local queue loads initialized near saturation limits (90%). It implements Interaction 43 cascades, meaning the battery collapse of an asset instantly escalates the computational load over surviving neighbors.

2.7 The Non-Convex Ground Obstructions

Planar 2D configuration-space keep-out zones embedded in the 100-mile grid, consisting of a Mountain Range (inducing wireless signal occlusion) and an Urban City Core (inducing bottleneck constraints).

2.8 The Battery Life & Autonomous Recharging Hubs

Non-linear power tracking variables coupled with an active infrastructure station at coordinates $[50.0, 50.0]$. If a platform's battery drops below a 30% threshold, the LISP DSL overwrites its tracking vector to execute an automatic battery swap, neutralizing Interaction 1 deletion cascades.

2.9 The Discrete Morse Filtration Engine

A combinatorial topological engine that maps a non-degenerate height filtration field $f : K \rightarrow \mathbb{R}$ over the graph complex. By evaluating face/co-face inequalities ($f(\alpha) \leq f(v)$), it pinpoints the exact coordinates of Critical Minima Traps (c_0) and Critical Saddle Edge Tunnels (c_1) like the open path $(0, 2)$.

2.10 The Self-Similar Group Controller

An algebraic scheduler governed by a Symmetric Group S_3 action. It converts spatial paths through the Morse corridors into generator words ($g_1 \cdot g_2 \cdot g_1$). Its recursive identity laws allow instantaneous inverses, supporting transactional, loss-free data rollbacks during edge-snapping events.

2.11 The Ground Rescue Vehicle

The final mechanical extraction asset. Once the store-and-forward computational buffers flush the verified visual sensor coordinates back to dispatch, the ground vehicle is targeted to the coordinates, achieving complete structural closure of the Attainable Space (V').